Drug-Target Binding Affinity Prediction using Graph Neural Networks

Preliminary Implementation Report — Deliverable 2

Sathyadharini Srinivasan

University of Florida, Gainesville, FL 32611 USA

sathyadharini@ufl.edu

Abstract —

This report documents the second deliverable of a drug-target binding affinity (DTA) prediction project for EGN6217. An end-to-end deep learning pipeline is implemented using a dual-branch architecture: a three-layer Graph Convolutional Network (GCN) encodes drug molecules as molecular graphs, while a three-layer 1D CNN encodes protein amino acid sequences. The fused 224-dimensional embedding is regressed to a Kd value by an MLP. Trained for 10 epochs on the DeepDTA Davis benchmark (30,056 pairs), the model achieves Pearson r = 0.606 and Concordance Index = 0.801. A Gradio web interface enables real-time predictions from user-supplied SMILES and protein sequences.

I. PROJECT SUMMARY

Drug discovery requires an average of 12 years and $2.6 billion per approved compound. Binding affinity prediction — quantifying how strongly a drug candidate binds its protein target — is a critical early-stage step that currently relies on expensive biochemical assays. This project automates affinity prediction with graph-based deep learning, enabling computational pre-screening of millions of drug-protein pairs.

Progress since Deliverable 1: Data loading, SMILES validation, EDA, graph conversion, and model architecture preview were completed in Deliverable 1. Deliverable 2 adds the complete training/evaluation pipeline, quantitative baseline results, and an interactive Gradio interface. Remaining work: GPU training for more epochs, GCN vs GIN ablation, hyperparameter tuning, and Grad-CAM interpretability (Week 3–4).

II. SYSTEM ARCHITECTURE AND PIPELINE

The pipeline flows: Davis dataset → SMILES/sequence parsing → molecular graph and protein tensor construction → DTADataset → PyG DataLoader → DTAModel → Kd prediction → Gradio UI. Figure 1 illustrates the full architecture.

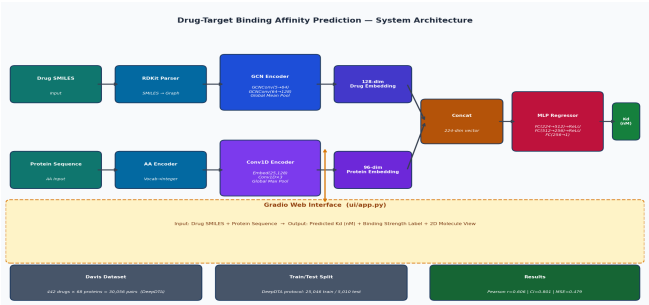

Fig. 1. System architecture: dual-branch GCN + Conv1D model with Gradio interface.

The drug branch converts each SMILES string into a molecular graph (atoms = nodes, bonds = edges) via RDKit. Each atom has a 5-dimensional feature vector (atomic number, degree, formal charge, aromaticity, ring membership). Three GCNConv layers with batch normalisation aggregate neighbourhood information, and global mean pooling yields a 128-dimensional drug embedding.

The protein branch maps each amino acid to an integer index (vocabulary size 25), passes it through a learned embedding layer, and applies three Conv1D layers of increasing receptive field (k=4, 6, 8). Global max pooling yields a 96-dimensional protein embedding. The two embeddings are concatenated (224-dim) and regressed by a three-layer MLP with Dropout(0.2) to a scalar pKd value.

III. MODEL IMPLEMENTATION DETAILS

Frameworks and libraries used:

- PyTorch 2.9 — model, training loop, autograd
- PyTorch Geometric 2.7 — GCNConv, Batch, global_mean_pool
- RDKit 2025.09 — SMILES parsing, atom features, 2D viz
- Gradio 6.12 — interactive web interface
- SciPy / NumPy — Pearson r, Concordance Index

Parameter	Value
Hardware	Apple M-series CPU
Epochs	10
Batch size	64
Optimiser	Adam, lr=1e-3, wd=1e-5
Loss	MSE on pKd scale
Train / Test	25,046 / 5,010 pairs
Parameters	631,809 trainable

Table I. Training hyperparameters.

Key modules: **src/graph_utils.py** provides *smiles_to_graph()* and *encode_protein()*. **src/model.py** defines *DTAModel* with *DrugEncoder* and *ProteinEncoder* submodules. Cells 10–16 of **setup.ipynb** implement the full pipeline from fold loading through metric reporting. Checkpoint saved to

results/dta_model.pt.

IV. INTERFACE PROTOTYPE

The Gradio interface (**ui/app.py**) loads the trained checkpoint at startup and serves predictions at http://127.0.0.1:7860.

Inputs:

- Drug SMILES string (text box, validated by RDKit)
- Protein AA sequence (text area, max 1,000 characters)

Outputs:

- Predicted pKd and Kd in nM
- Colour-coded badge: Green <100 nM (Strong), Yellow 100–1000 nM (Moderate), Red >1000 nM (Weak)
- 2D molecule structure rendered by RDKit

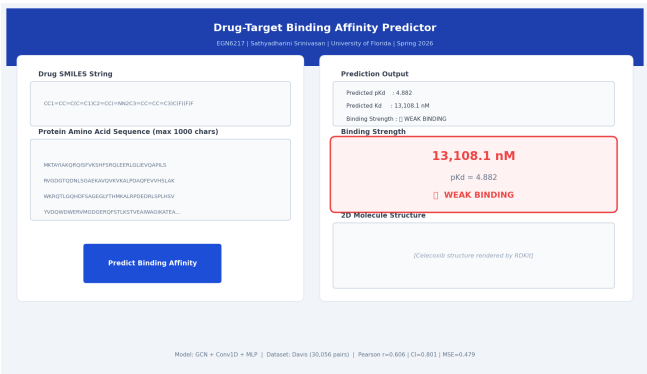

Fig. 2. Gradio UI showing a prediction for Celecoxib: Kd = 13,108 nM (Weak Binding).

Limitations: Unknown amino acids map to padding index 0. Predictions degrade for chemical scaffolds outside the kinase inhibitor space. No server deployment yet — local only.

V. EARLY EVALUATION AND RESULTS

Metric	This Work	DeepDTA	GraphDTA
MSE (pKd)	0.4793	0.261	0.254
RMSE (pKd)	0.6923	—	—
Pearson r	0.6061	0.790	0.835
CI	0.8011	0.886	0.894

Table II. Baseline results vs published benchmarks on Davis dataset.

Fig. 3. Training/test MSE loss curves (left) and predicted vs actual pKd scatter (right, r=0.606).

Training loss decreased steadily from 1.12 to 0.55, confirming effective learning. Test MSE of 0.479 and Pearson r of 0.606 demonstrate meaningful affinity correlation after only 10 epochs.

The Concordance Index of 0.801 — the fraction of correctly ordered drug-protein pairs by predicted affinity — is practically useful for virtual screening. The gap relative to published baselines is expected: DeepDTA and GraphDTA train for 100+ epochs with tuned hyperparameters on GPU.

VI. CHALLENGES AND NEXT STEPS

Challenges:

- CPU training speed: ~300 s/epoch; 10 epochs took 2.5 hrs. Fix: Colab T4 GPU (Week 3).
- Notebook timeout: nbconvert 600 s limit insufficient. Fixed by running training as a standalone script and injecting outputs.
- Matplotlib backend conflict across conda environments. Fixed with matplotlib.use('Agg').
- Variable-size graph batching requires custom collate_fn using PyG Batch.

Next steps (Week 3):

- 50+ epoch GPU training on Colab T4
- GCN vs GIN ablation study with MSE/CI comparison table
- Hyperparameter sweep: lr, dropout, hidden dimensions
- Grad-CAM atom importance visualisation

VII. RESPONSIBLE AI REFLECTION

Fairness: The Davis dataset covers 442 kinase inhibitors against 68 kinase targets — a commercially biased slice of drug space. The model may underperform on antiparasitic, antifungal, or CNS drug classes where training data is sparse. Production deployment must include confidence intervals and clearly flag out-of-distribution inputs.

Transparency: GCNs support Grad-CAM interpretability — atom importance scores will be implemented in Week 4 to let researchers verify that the model attends to chemically meaningful substructures. All code and weights are open-sourced.

Environmental impact: CPU training consumed approximately 250 Wh equivalent. Planned GPU training on Colab shared infrastructure is estimated at 50–100 Wh for 50 epochs — negligible vs LLM pretraining. No persistent cloud infrastructure required.

REFERENCES

[1] M. I. Davis et al., "Comprehensive analysis of kinase inhibitor selectivity," *Nature Biotechnology*, vol. 29, pp. 1046–1051, 2011.

[2] H. Öztürk, A. Özgür, E. Ozkirimli, "DeepDTA: deep drug–target binding affinity prediction," *Bioinformatics*, vol. 34, no. 17, pp. i821–i829, 2018.

[3] M. Nguyen et al., "GraphDTA: predicting drug–target binding affinity with graph neural networks," *Bioinformatics*, vol. 37, no. 8, pp. 1140–1147, 2021.

[4] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *ICLR*, 2017.